

Homework B1 | ECON 0100 | Fall 2026 | Solutions

Solution guide for the teaching team. Answers and working are in red; everything in black is what students see. Students answer on paper and submit selections on Gradescope, where the homework is completion-graded.

Butterbeer

Preferences for butter beer can be represented by the following demand curve:

$$P_d = 100 - \frac{1}{2}Q_d$$

Prices are in galleons and quantity is in bottles.

Q1 | Quantity Demanded

Use a graph to plot this demand curve, including the quantity demanded at both 40 galleons and 50 galleons.

- a) Quantity demanded at 40 galleons: 120 bottles
- b) Quantity demanded at 50 galleons: 100 bottles
- c) How much did quantity demanded change as the price dropped from 50 to 40? up by 20 bottles
- d) What is the marginal benefit at a quantity of 100? 50 galleons

Plot. Vertical intercept, $Q_d = 0$: $P_d = 100$. Horizontal intercept, $P_d = 0$: $0 = 100 - \frac{1}{2}Q_d \rightarrow \frac{1}{2}Q_d = 100 \rightarrow Q_d = 200$.

a) $40 = 100 - \frac{1}{2}Q_d \rightarrow \frac{1}{2}Q_d = 60 \rightarrow Q_d = 120$ b) $50 = 100 - \frac{1}{2}Q_d \rightarrow \frac{1}{2}Q_d = 50 \rightarrow Q_d = 100$

c) $\Delta Q_d = 120 - 100 = 20$: quantity demanded rose by 20 bottles as the price fell, the law of demand.

d) Quantity to price. The height of demand at 100 bottles is the marginal benefit of the hundredth bottle: $P_d = 100 - \frac{1}{2} \cdot 100 = 50$ galleons. It matches b): 50 galleons is the price at which 100 bottles are demanded.

Q2 | Consumer Surplus

Then find and label the consumer surplus at these prices.

- a) CS at 40 galleons: 3600 galleons
- b) CS at 50 galleons: 2500 galleons
- c) How much did consumer surplus change as the price dropped from 50 to 40? up by 1100 galleons

Consumer surplus is the triangle below demand and above the price, out to the quantity bought from Q1. Its height is the 100-galleon intercept minus the price and its base is the quantity.

$$P = 40 : \quad CS = h \cdot b \cdot \frac{1}{2} = (100 - 40) \cdot 120 \cdot \frac{1}{2} = 60 \cdot 60 = 3600$$

$$P = 50 : \quad CS' = h \cdot b \cdot \frac{1}{2} = (100 - 50) \cdot 100 \cdot \frac{1}{2} = 50 \cdot 50 = 2500$$

c) $\Delta CS = CS - CS' = 3600 - 2500 = 1100$: consumer surplus rose by 1100 galleons, the strip between the two price lines. Expenditure is the rectangle below the price: $40 \cdot 120 = 4800$ at 40 galleons and $50 \cdot 100 = 5000$ at 50.

